TRUSS MASTER

Bridge Building Competition

Mecha Rush'26 · Dept. of Mechanical Engineering · BSACIST · April 7, 2026

EVENT BRIEF

A structural engineering competition that challenges teams to design and build the most efficient truss bridge possible; by applying principles of load distribution, structural mechanics, and material optimization under real time and resource constraints.

Your bridge will be judged not just on how much it holds, but on how efficiently it holds it. Precision, planning, and smart material use are what separate the best structures from the rest.

VENUE & TENTATIVE SCHEDULE

Venue Mechanical Science Block, 1st Floor - Back Side (Opp. Robo Lab)

Schedule April 7, 2026 | Full-day Event → 10.30AM - 1PM (Construction) & After 1.30PM (Evaluation)

TEAMS

- 3 to 4 members per team
- Cross-department teams permitted; cross-college teams are not allowed
- A participant cannot be part of more than one team
- All participants must carry their college ID and conformation mail

BRIDGE SPECIFICATIONS

Length	25 – 35 cm
Height	15 – 20 cm
Breadth	10 – 15 cm

Bridges that do not conform to the dimension limits above will be eliminated before evaluation.

RULES & REGULATIONS

- All construction of the bridge must be completed on-site within the allotted building time by only using the materials distributed by the organizers.
- No adhesives, reinforcements, or external materials are permitted.
- Measuring, marking, and cutting tools must be brought by participants if required.
- Design and dimensions must strictly follow the technical specifications above.
- Use of electronic gadgets during the build phase is not permitted.
- The final structure will be subjected to a load-bearing test to determine efficiency.
- Jury decisions are based on structural integrity, build time, and final composite score.
- The organising committee has the right to modify rules, participants will get informed if changes made

EVALUATION

Final Score = **Weighted composite across four criteria (total 100%)**

Criterion	Weightage	Basis
Structural Integrity	**10%**	*Professional truss links; aesthetics included*
Strength-to-Weight Ratio	**60%**	*Team percentile based on load vs bridge weight*
Build Time	**20%**	*Team percentile based on time to complete construction*
Material Efficiency	**10%**	*Team percentile: bridge weight vs raw material weight*

How is the Strength-to-Weight Ratio scored?

Each team's load-bearing result is ranked against all other teams. Your percentile rank is then converted to a score out of 60. A bridge that holds the most relative to its own weight scores highest — raw load alone is not enough.

DISQUALIFICATION

- Bridge dimensions fall outside the specified limits
- Use of materials or adhesives not provided by the organizers
- Any external assistance or modification after the build phase ends
- Disruption of the event or tampering with another team's structure
- Violation of any rule at the discretion of the jury

Judges' decisions are final. Event coordinators reserve the right to modify rules for smooth execution.

For queries, contact the organizing committee through contact@mecharush.in | mecharush.in